

10. Regulation of a three phase Alternator by emf and mmf methods

Objectives:

Predetermine the regulation of alternator by emf and mmf methods at 0.8pf lag, upf and 0.8pf lead.

11. Regulation of a three phase alternator by Potier method

Objectives:

- a) Synchronize the alternator by dark lamp method
- b) Plot ZPF characteristics and determine armature reactance mmf and potier reactance
- c) Predetermine the regulation by ZPF method

12. Reactive power control in grid connected Alternators

Objectives:

- a) Synchronize the alternator by bright lamp method
- b) Control the reactive power and plot the V and inverted V curves for generator operation

13. Slip Test on a three phase Salient Pole Alternator

Objectives:

- a) Determine the direct and quadrature axis synchronous reactances
- b) Predetermine the regulation at 0.8 lagging power factor

14. V/f control of three phase Squirrel Cage Induction Motor

Objectives:

Perform speed control of the given three phase induction motor by V/f control

15. Performance characteristics of a three phase Induction Generator

Objectives:

Plot the performance characteristics of the generator.